Geospatial Risk-Adjusted Housing Intelligence Using Public Data and Machine Learning

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1 Project Overview

Housing affordability and short-term housing value trends are shaped by a combination of market momentum, local socioeconomic conditions, environmental exposure, and hazard-related context. While housing price data is widely available, these other factors are often fragmented across public sources and difficult to interpret in a single decision-making framework. This project addresses that problem by building a ZIP-level housing trend prediction tool for Pennsylvania, where the author recently moved, that integrates housing, demographic, environmental, and flood-related data into one predictive modeling workflow.

The final deliverable is a data-driven decision-support system that predicts the 12-month percent change in single-family home values for Pennsylvania ZIP codes. The system combines Zillow Home Value Index (ZHVI) data, American Community Survey (ACS) demographic variables, EPA Toxics Release Inventory (TRI) environmental context, and FEMA flood-related data. The results are presented in a working Streamlit application that allows the user to select a ZIP code and explore the predicted housing trend, actual observed trend, and simplified environmental context. The project is highly customizable to include all states in the United States, but is presented here for Pennsylvania only, for the sake of brevity.

2 Project Goals

The primary goal of this project is to determine whether recent housing market behavior and selected geographic context variables can be used to predict ZIP-level housing value trends in Pennsylvania.

Specific goals of the project were to:

- build a reproducible ZIP-level modeling dataset;
- define a target variable based on 12-month percent change in single-family Zillow Home Value Index;
- integrate socioeconomic variables from ACS;
- incorporate environmental exposure and hazard context using EPA TRI and FEMA flood data;
- compare baseline and more advanced machine learning approaches;
- select the best-performing model based on predictive performance;
- and deploy the final model in a user-facing Streamlit application.

Research Question:

Can recent housing trends, demographic conditions, and environmental/hazard context be used to predict 12-month ZIP-level housing value change in Pennsylvania?

3 Data Sources

This project integrates four primary public datasets.

3.1 Zillow Home Value Index (ZHVI)

The Zillow Single-Family ZHVI ZIP-level dataset was used as the core housing data source. The full nationwide file was filtered to Pennsylvania ZIP codes only. This dataset was used to construct the target variable and several recent housing momentum features.

3.2 American Community Survey (ACS)

ACS ZIP Code Tabulation Area (ZCTA)-level data was used to add demographic and socioeconomic context. The following ACS tables were used:

- **B19013** — Median household income
- **B15003** — Educational attainment
- **B25003** — Owner-occupied versus renter-occupied housing
- **B01001** — Total population

These files were transformed into ZIP-level features and merged into the Zillow modeling dataset.

3.3 EPA Toxics Release Inventory (TRI)

EPA TRI data was used to engineer county-level environmental exposure features. These included:

- the number of TRI-reporting industrial facilities in each county
- the total annual reported chemical releases in each county

These environmental exposure measures were then merged back into the ZIP-level housing data using county relationships.

3.4 FEMA Flood Data

FEMA flood-hazard files were used to create a simplified county-level flood hazard proxy. Flood-hazard polygon counts were derived from selected Pennsylvania county flood files and merged into the ZIP-level model as an additional hazard-related contextual feature.

4 Target Variable

The target variable for the project is the 12-month percent change in Single-Family Zillow Home Value Index (ZHVI) ending in March 2026.

$$y = \frac{ZHVI_{2026-03} - ZHVI_{2025-03}}{ZHVI_{2025-03}} \times 100$$

This target was chosen because the approved proposal emphasized the prediction of housing price trends, not just static home values.

5 Feature Engineering

The final modeling table combines housing trend features, demographic context variables, environmental exposure metrics, and hazard-related context.

5.1 Housing Features (Zillow)

The Zillow-derived features included:

- zhvi_2026_03
- zhvi_2025_03
- pct_chg_1m
- pct_chg_3m
- pct_chg_6m
- vol_12m

These variables capture recent price behavior and short-term volatility.

5.2 Socioeconomic Features (ACS)

The ACS-derived features included:

- median_household_income
- pct_bachelor_or_higher
- pct_owner_occupied
- population_total

5.3 Environmental and Hazard Features

Added contextual variables included:

- tri_facility_count_county
- tri_total_releases_county
- flood_zone_count

These features were merged at the county level to provide a simplified environmental and flood-related context for each ZIP.

6 Modeling Approach

The project used a supervised regression framework to predict 12-month ZIP-level housing trend change.

6.1 Baseline Model

A Linear Regression model was used as the baseline because it is simple, interpretable, and easy to explain.

6.2 Advanced Model

A Random Forest Regressor was used as a more advanced nonlinear comparison model.

6.3 Data Preparation

The dataset was divided into training and test subsets using an 80/20 split. Missing numeric values were handled using median imputation within a preprocessing pipeline.

6.4 Evaluation Metrics

Model performance was evaluated using:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2

7 Model Results

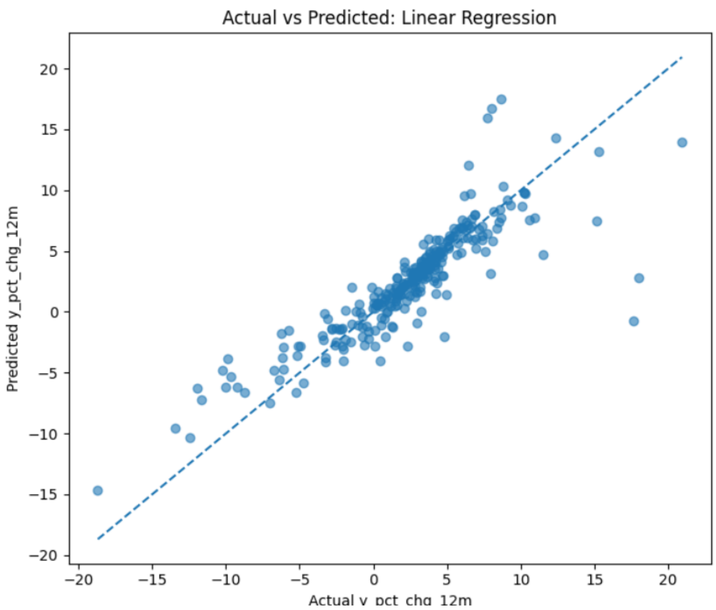
Three final model versions were compared:

1. Linear Regression (Zillow + ACS)
2. Linear Regression (Zillow + ACS + TRI)
3. Linear Regression (Zillow + ACS + TRI + FEMA)

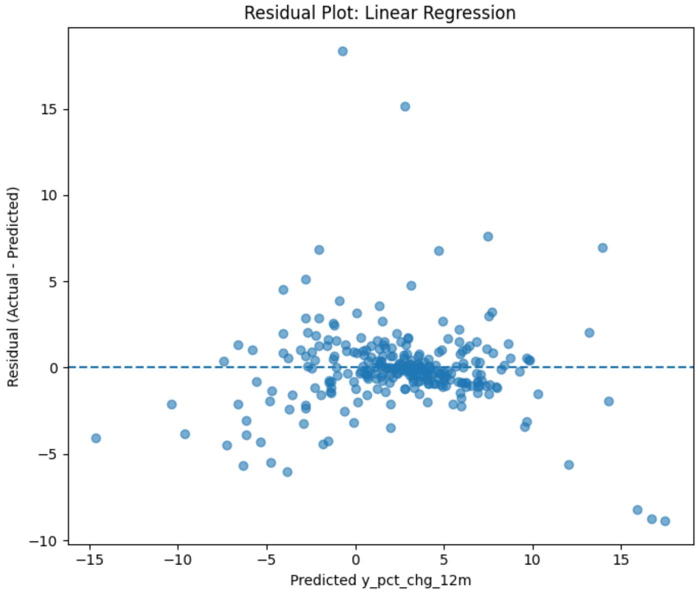
7.1 Comparison Table

Model Version	MAE	RMSE	R^2
Linear Regression (Zillow + ACS)	1.4333	2.5018	0.7480
Linear Regression (Zillow + ACS + TRI)	1.4413	2.5008	0.7482
Linear Regression (Zillow + ACS + TRI + FEMA)	1.4405	2.5014	0.7481

The baseline Zillow + ACS model had the lowest MAE, while the TRI-enhanced version showed a very slight improvement in RMSE and R^2. Adding the flood feature produced only a small additional performance change. Overall, environmental and hazard variables did not materially improve predictive accuracy, but they did enhance the project's environmental and geospatial risk framing.



Actual vs Predicted Plot



Residual Plot

8 Model Interpretation

The strongest predictive signals in the final linear model came primarily from recent housing trend and volatility features. The most influential variables included:

- vol_12m
- pct_chg_6m
- pct_chg_1m
- pct_chg_3m
- pct_owner_occupied

Recent housing momentum features carried the most predictive value, while ACS, TRI, and FEMA variables contributed additional contextual information rather than dramatic predictive lift. This is still an important result because it shows that environmental and hazard information can be integrated successfully into a housing trend model even when the strongest predictive drivers remain market-based features.

9 Example ZIP: 18109

ZIP code 18109 was used as the primary demonstration ZIP for the final application.

For ZIP 18109:

- Actual 12-month % change: 5.37%
- Predicted 12-month % change: 5.86%

Environmental and hazard context for ZIP 18109 included:

- Exposure score: High
- Industrial facilities reporting (county): 21
- Industrial chemical releases (county, annual): ~1.07 million pounds
- Mapped flood hazard zones (county): 1254

This ZIP-level example demonstrates how model predictions and simplified risk context can be shown together in a practical decision-support tool.

Pennsylvania ZIP Housing Trend Predictor

This app predicts 12-month % change in single-family Zillow Home Value Index (ZHVI) for Pennsylvania ZIP codes using housing, socioeconomic, environmental, and flood-related features.

Select a Pennsylvania ZIP code

18109

Prediction

Predicted 12-month % change

5.86%

Actual 12-month % change

5.37%

ZIP Details

ZIP: 18109

County: Lehigh County

Metro: Allentown-Bethlehem-Easton, PA-NJ

Environmental Context

Exposure score (county): High

Industrial facilities reporting (county): 21

Industrial chemical releases (county, annual): ~1.07 million pounds

Mapped flood hazard zones (county): 1254

This app uses county-level EPA TRI and FEMA-derived flood context as simplified environmental and hazard indicators. These values provide context rather than direct property-level risk estimates.

Summary

For ZIP **18109**, the model predicts a **5.86%** change in single-family ZHVI over 12 months.

Feature Values

	feature	value
0	zhvi_2026_03	265973.0485
1	zhvi_2025_03	252408.9571
2	pct_chg_1m	0.1549
3	pct_chg_3m	0.794
4	pct_chg_6m	2.5002
5	vol_12m	0.1494
6	median_household_income	58031
7	pct_bachelor_or_higher	15.9742
8	pct_owner_occupied	52.0768
9	population_total	18446

10 Streamlit Application

The final Streamlit application allows the user to:

- select a Pennsylvania ZIP code;
- view the predicted 12-month ZHVI percent change;
- compare the predicted value to the actual observed value;
- view ZIP details such as county and metro;
- inspect the model input feature values;
- and review environmental context, including an exposure score and county-level flood-hazard count.

The application operationalizes the model as a simple, user-facing dashboard consistent with the project's original goal of producing an interpretable decision-support system.

11 Limitations

This project has several limitations:

1. **County-level aggregation**
   EPA TRI and FEMA-related variables were merged at the county level rather than the parcel or ZIP polygon level.

2. **Partial FEMA coverage**
   The flood-related feature was based on selected Pennsylvania counties rather than all 67 counties.

3. **Simplified environmental interpretation**
   TRI and FEMA variables are used as contextual indicators rather than direct property-level risk measures.

4. **Limited feature set**
   The project does not include all possible local economic or housing drivers, such as mortgage rates, labor-market measures, or parcel-level flood boundaries.

These limitations are acknowledged, but they do not undermine the overall value of the project as an end-to-end applied data science workflow.

12 Conclusion

This project successfully developed a Pennsylvania ZIP Housing Trend Predictor that integrates:

- Zillow housing trends,
- ACS socioeconomic features,
- EPA TRI environmental context,
- FEMA flood-related hazard context,
- regression-based predictive modeling,
- and a working Streamlit application.

The final result is a functional, interpretable, and proposal-aligned applied project. While environmental and hazard variables added only modest predictive improvement, they strengthened the project's geospatial risk framing and improved the contextual value of the final decision-support tool.

13 Use of Generative AI

Portions of this document were developed with the assistance of Microsoft Copilot to support drafting, organization, and clarity of written content. All conceptual decisions, analytical methods, and final edits are the responsibility of the author.